

PersonalQuery: Reproducibility Package for Grounded Personalized Query Generation

Anonymous
Anonymous Institution
anonymous@email.org

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Abstract

This reproducibility companion presents PersonalQuery, a complete artifact package for grounded personalized query generation. The original work introduced a 12-stage methodology achieving +1.81 points improvement (45% relative gain). Our reproducibility package includes: (1) complete 12-stage pipeline; (2) benchmark dataset with 134 paired queries across 10 users; (3) comprehensive documentation; (4) single-command reproduction scripts; and (5) artifacts publicly archived on Zenodo with DOI. The package enables full reproduction within 15-30 minutes per user with expected outputs matching paper claims within ± 0.2 tolerance. We target ACM badges: Artifacts Available, Artifacts Evaluated-Functional, and Results Reproduced.

1 Introduction

Personalized search query generation represents an emerging challenge in recommender systems research. The original PersonalQuery work [Anonymous, 2026] introduced a novel grounded methodology extracting verified user preferences from reviews and generating queries reflecting individual writing styles.

This companion focuses on reproducibility, providing a complete artifact package enabling researchers to reproduce all main experimental results with single commands.

2 Artifact Overview

Claim 1: Personalized queries significantly outperform public queries.

Reproduction: Script `reproduce_main_results.py` evaluates query pairs and outputs scores matching Table 2 (expected: Public avg ≈ 4.0 , Personalized avg ≈ 5.8 , improvement $\approx +1.81$).

Claim 2: Grounded validation filters 71.5% of preferences.

Reproduction: Pipeline execution shows preference counts before/after validation.

Claim 3: 100% of users show positive improvement.

Reproduction: Full evaluation script outputs per-user improvements, all expected positive.

3 Reproduction Methodology

Installation (5-10 minutes):

```
git clone https://github.com/[user]/personalquery.git
cd personalquery
conda env create -f environment.yml
conda activate personalquery
python scripts/verify_setup.py
```

Main Results (15-30 minutes):

```
python scripts/reproduce_main_results.py --user-id A130FOB1394G31
```

Expected: Public 4.0, Personalized 5.8, Improvement +1.81 (± 0.2 tolerance).

Full Evaluation (2-4 hours):

```
python scripts/run_full_evaluation.py --all-users
```

4 Technical Requirements

Hardware: CPU 8+ cores, RAM 16GB minimum, Storage 5GB.

Software: Python 3.8+, conda/pip, LLM API (GLM-4.5-Air or OpenAI GPT-4).

Dependencies: openai, spacy, sentence-transformers, pandas, numpy, scikit-learn.

5 Artifact Availability

Primary: Zenodo (<https://zenodo.org/record/XXXXXX>)

Secondary: GitHub ([https://github.com/\[username\]/personalquery](https://github.com/[username]/personalquery))

License: MIT (allows comparison and extension)

References

Anonymous. Personalquery: A benchmark dataset for personalized search query generation. In *ACM Conference on Recommender Systems (RecSys)*, 2026.